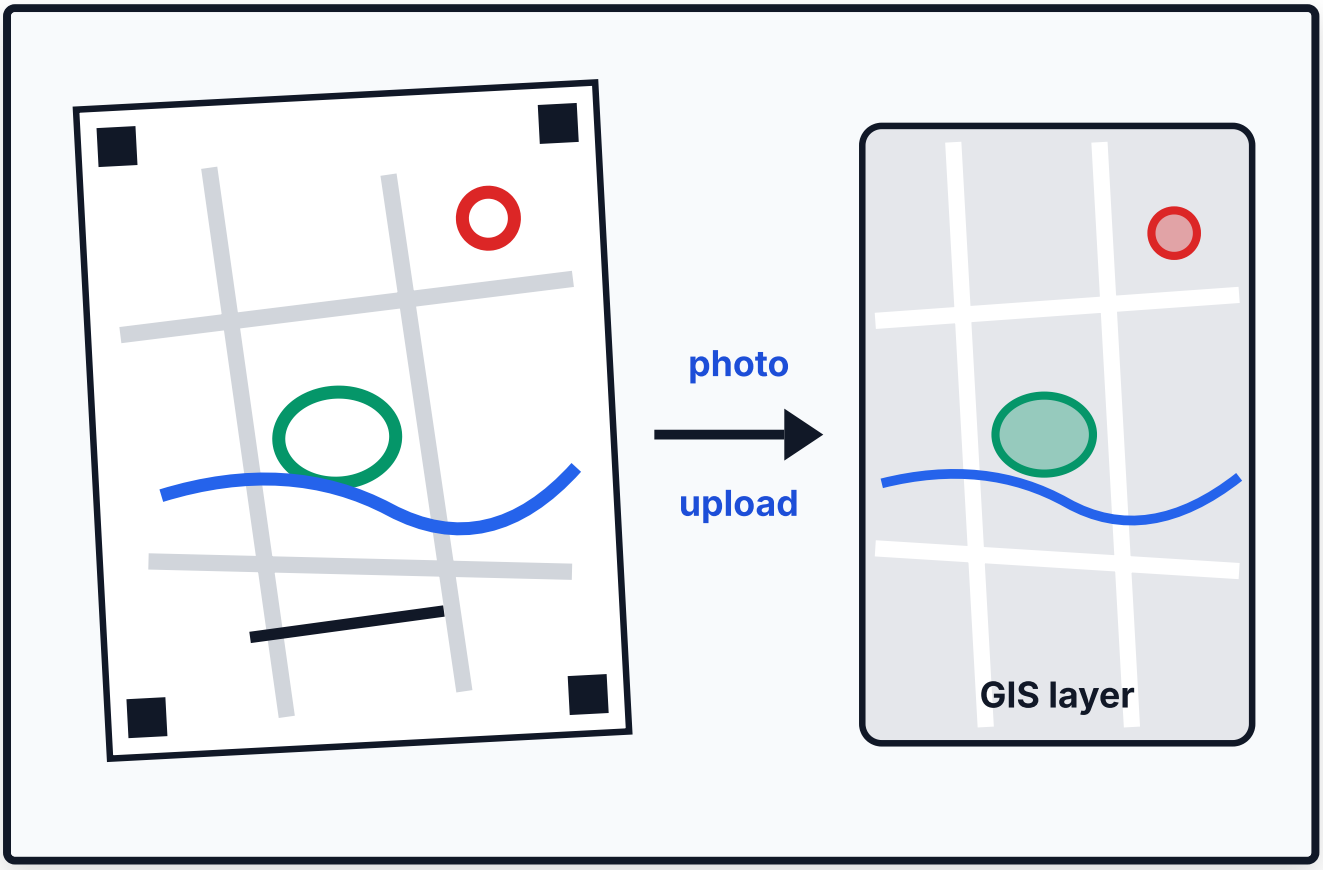
MAPPING WITH KIDS: A MAPPING ACTIVITY

From Paper to Map

Become a neighbourhood mapper: draw on a paper map, then turn your drawing into real GIS data



Your drawing on paper becomes a layer you can open in GIS software.

What you'll use

- **Sketch Map Tool**: a free website that makes paper maps and reads your drawings back (sketch-map-tool.heigit.org)
- **A printer** and A4 or A3 paper
- **Thick felt-tip markers** in red, blue, green, and black
- **A phone camera or a scanner**
- **GeoLibre**: for the last step, a free, web-based map-making tool

You Are Already a Mapmaker

You know things about your neighbourhood that no satellite can see: the shortcut to school, the tree with the best shade, the corner that floods every monsoon, the shop that sells the best snacks.

Today you'll put that knowledge on a map. You'll draw on **paper** with **markers**, just like an ordinary drawing. Then a computer will read your drawing and turn it into **real GIS data**, placed exactly where it belongs on Earth.

> **GIS WORD OF THE DAY**
> **Sketch map**: a map drawn by hand by someone who knows a place well. Sketch maps show what *people* know, not just what a satellite photographs.

> **FUN FACT**
> The Sketch Map Tool was made by HeiGIT, a research institute in Heidelberg, Germany. Communities around the world use it to mark places at risk from floods and other disasters, so that help can reach them faster.

Here's the plan for today, in six steps:

1. Make your paper map
2. Plan your colour key
3. Draw on your map
4. Take a good photo
5. Upload it and let the computer read it
6. Open your drawing in GeoLibre

1 Make Your Paper Map

First, ask the Sketch Map Tool to make a map of your area, ready to print.

1. Go to **sketch-map-tool.heigit.org** and choose **Create** to make a new sketch map.
2. Search for your school or your neighbourhood, then zoom and move the map until it shows the area you want. Keep it **small**: a few streets around your school is perfect.
3. Under **Page Setup**, choose the **Paper Format** (A4 if you're mapping alone, A3 if you're mapping in a group) and the **Paper Orientation**.
4. Choose a street map or a satellite base map. Street maps are easier to draw on.
5. Click **NEXT**, then download the PDF and print it.

> **TIP**
> Look at the edges of your printed map. There are special patterns there. They tell the computer exactly which part of the Earth your map shows, so **never draw over them** and never cut them off.

> **FUN FACT**
> The street map comes from OpenStreetMap, a map of the whole world drawn by volunteers. If a street near you is missing, you could become an OpenStreetMap mapper too!

2 Plan Your Colour Key

Before drawing, decide what each colour means. Every real map has a key, and yours will too. Here's an example. You can change the meanings, but keep these four colours, because the computer reads them best.

Colour	What it means on my map
Red	My favourite places (where I play, eat, or meet friends)
Blue	Water: lakes, drains, and spots that flood when it rains
Green	Trees, parks, and green spaces
Black	The path I walk or ride to school

Write your own key here:

Colour	What it means on my map
Red	
Blue	
Green	
Black	

GIS WORD OF THE DAY

Legend (or key): the part of a map that explains what each colour and symbol means.

3 Draw on Your Map

Now the fun part. Follow these drawing rules so the computer can read your map.

1. Use **thick felt-tip markers** with strong colours. Thin, light lines are hard to read.
2. To mark an **area** (like a park), draw a **closed outline** with no gaps. Don't colour it in.
3. To mark a **place** (like a shop), draw a **big circle** around it, not a tiny dot.
4. To mark a **route** (like your path to school), draw one clear line.
5. Keep away from the patterns on the edges of the map.

> **TIP**
> Smaller areas are read better than very big ones. If you want to mark a huge area, try splitting it into a few smaller shapes.

> **KEEP IT SAFE**
> Mark your **school** and **public places**, but don't mark your own home. Maps can be shared, and your home address is private.

> **GIS WORD OF THE DAY**
> **Point, line, polygon**: the three shapes of GIS. A shop is a *point*, a road is a *line*, and a park is a *polygon* (a closed shape).

4 Take a Good Photo

The computer only sees what your photo shows, so a good photo matters a lot.

1. Lay the map **flat** on a table. Smooth out any folds.
2. Find good, even light. Avoid shadows from your hand or phone falling on the map.
3. Hold the phone **straight above** the map, flat and parallel to the paper, not at a slant.
4. Make sure the **whole sheet**, including all the edge patterns, is in the photo.
5. Check the photo is sharp before moving on. If it's blurry, take another one.

> **TIP**
> A scanner works even better than a phone. If you have one, scan the map at 300 dpi and save it as a JPG or PNG.

> **FUN FACT**
> Satellites have the same problem as your phone! Clouds, shadows, and slanted angles make satellite photos harder to use, so scientists work hard to correct them.

5 Upload and Let the Computer Read It

Now the computer turns your drawing into data.

1. On the Sketch Map Tool website, choose **Digitize**.
2. Click **Choose file(s)** and pick your photo, or drag it onto the page.
3. The tool asks if you want to share your data to help improve it. Ask your teacher or parent which button to press: **Agree & Upload** or **Reject & Upload**.
4. Wait while the tool works. It finds your map, lines it up with the real world, and picks out each colour you drew.
5. Download your results. You'll get a **GeoJSON** file (your drawings as shapes) and a **GeoTIFF** file (your photo, placed on the Earth).

GIS WORD OF THE DAY

Georeferencing: matching a picture to real places on Earth, so every spot on the picture gets a latitude and longitude. The edge patterns on your map make this possible.

TIP

If some of your drawings are missing, the photo was probably too dark or the lines too thin. Go over them with a thicker marker and try again.

6 Open Your Drawing in GeoLibre

Your hand drawing is now real GIS data. Let's open it in a GIS tool.

1. Open GeoLibre and click **Launch GeoLibre Web**.
2. Add your GeoJSON file with **Add Data** → **Vector Layer**.
3. Your shapes appear, in the right place on the map! Zoom in to your neighbourhood.
4. Open the **Style** panel and colour the shapes using the `color` column, so red drawings show in red, blue in blue, and so on.
5. Switch to a **satellite** base map. Is the park you drew really green from space? Is the lake where you drew it?

> **TIP**
> You can also add the GeoTIFF with **Add Data** to see your whole photo laid over the real map. Turn it on and off to compare your drawing with the satellite view.

> **FUN FACT**
> City planners use maps like yours to plan parks, find flood-prone streets, and decide where new schools should go.

You Made Real GIS Data!

You started with a pen and paper and ended with data that any GIS software in the world can open. You also added something no satellite knows: what *you* know about your place.

What You Learned

- What a **sketch map** is and why local knowledge matters
- Making a map **legend** (colour key)
- The three GIS shapes: **point, line, polygon**
- How a photo becomes data through **georeferencing**
- Two common GIS file types: **GeoJSON** and **GeoTIFF**
- Opening and styling your own data in **GeoLibre**

Try It Yourself

- Ask a grandparent or neighbour to draw on a copy of the same map. What do they know that you didn't?
- Map the same streets in the dry season and again after heavy rain. Where does the water collect?
- Put everyone's maps from your class into GeoLibre together. Which places did lots of people mark?

FOR TEACHERS AND PARENTS

Print the maps before class and test your markers on one sheet first. Check that OpenStreetMap shows the streets in your area well. Children should mark public places only, not homes. On upload, decide in advance whether to share the drawings with the tool's researchers (Agree) or not (Reject).